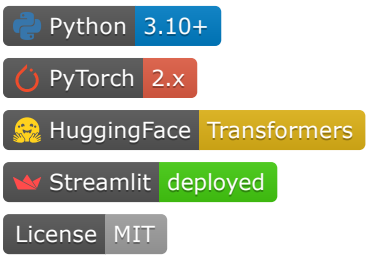
AI-Powered Movie Review Movie Analyzer

> An end-to-end deep learning NLP project that classifies movie reviews as **positive** or **negative** using a fine-tuned DistilBERT transformer model which is deployed as a live interactive web app.



Overview

This project demonstrates a complete machine learning pipeline, starting from raw data exploration ending in a deployed web application. It uses the IMDB dataset (50,000 movie reviews) and fine-tunes a **DistilBERT** model to classify sentiment with ~93% accuracy.

Live Demo

Try it on Hugging Face Spaces

Results

Metric	Score
Accuracy	93.2%
F1 Score	0.932
Precision	0.934
Recall	0.930

Evaluated on 25,000 held-out IMDB test reviews.

Project Structure

```
movie_classifier/
│
├── data/
│   └── raw/                    # Original IMDB dataset
│
├── notebooks/
```

```
│   ├── 01_EDA.ipynb            # Exploratory data analysis
│   └── 02_training.ipynb       # Model training walkthrough
│
├── src/
│   ├── preprocess.py           # Tokenization & data loading
│   ├── train.py                # Training loop & config
│   ├── evaluate.py             # Metrics & confusion matrix
│   └── predict.py              # Inference on new reviews
│
├── app/
│   └── app.py                  # Streamlit web application
│
├── models/
│   └── best_model.pt           # Saved fine-tuned model
│
├── assets/
│   └── demo_screenshot.png     # App screenshot for README
│
├── requirements.txt
└── README.md
```

Tech Stack

Layer	Tool / Library
Language	Python 3.10+
Deep Learning	PyTorch 2.x
NLP Model	DistilBERT (HuggingFace)
Data	HuggingFace `datasets`
Web App	Streamlit
Deployment	Hugging Face Spaces
Experiment Tracking	(optional) Weights & Biases

How to Run Locally

1. Clone the repository

```
git clone https://github.com/johannesfelzmann/movie_classifier.git
cd movie_classifier
```

2. Create a virtual environment

```
python -m venv venv
source venv/bin/activate        # On Windows: venv\Scripts\activate
```

3. Install dependencies

```
pip install -r requirements.txt
```

4. Train the model

```
python src/train.py
```

> Training takes ~20 minutes on a GPU, ~2 hours on CPU.
> A pre-trained checkpoint is available in `/models/best_model.pt`.

5. Launch the web app

```
streamlit run app/app.py
```

Open http://localhost:8501 in your browser.

Model Architecture

This project fine-tunes **DistilBERT** (`distilbert-base-uncased`), a lightweight version of BERT that retains 97% of BERT's language understanding while being 40% smaller and 60% faster.

```
Input Review Text
       ↓
DistilBERT Tokenizer (max 512 tokens)
       ↓
DistilBERT Encoder (6 transformer layers)
       ↓
[CLS] token representation
       ↓
Linear Classification Head (768 → 2)
       ↓
Softmax → Positive / Negative
```

Training config:

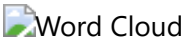
- Optimizer: AdamW

- Learning rate: 2e-5 with linear warm-up
- Batch size: 16
- Epochs: 3
- Loss: CrossEntropyLoss

Exploratory Data Analysis

Key findings from `notebooks/01_EDA.ipynb`:

- Dataset is **perfectly balanced** (25k positive, 25k negative)
- Average review length: **~230 words**
- Most common positive words: *great, excellent, wonderful, best, amazing*
- Most common negative words: *bad, worst, waste, boring, awful*



Data

The project uses the IMDB Large Movie Review Dataset:

- **50,000** labeled movie reviews
- **25,000** for training, **25,000** for testing
- Binary labels: positive (1) / negative (0)

To download via HuggingFace:

```
from datasets import load_dataset
dataset = load_dataset("imdb")
```

Future Improvements

- [ ] Add **confidence score** display in the UI
- [ ] Support **batch predictions** from CSV upload
- [ ] Track experiments with **Weights & Biases**
- [ ] Store prediction logs in **PostgreSQL** database
- [ ] Fine-tune on domain-specific data (product reviews, tweets)
- [ ] Add **explainability** with SHAP or attention visualization

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